

A Lean-Verified Single-Transition Theorem for CMV Ball Density

Proof built, Lean-verified, and semantically accepted by
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The candidate comparison

Cañete, Miranda Jr., and Vittone study the planar isoperimetric problem with density

[$f(x) =$

$$\begin{cases} \lambda, & |x| \leq 1, \\ 1, & |x| > 1, \end{cases}$$

< 1 .]

Weighted area is the integral of this density over a region; weighted perimeter uses the same density on its boundary. In particular, an arc on the unit circle has weight λ . The relevant candidates are regions of type (B), bounded by an arc of the unit circle and an exterior circular arc meeting it at angle θ , and balls of type (C), whose boundaries meet the unit circle orthogonally. The source classification and candidate formulas are in [CMV, Proposition 3.22, pp. 32–33, and Theorem 3.23, pp. 34–36]. We prove an analytic comparison of these two complete candidate families. Its application to arbitrary finite-perimeter minimizers requires the source-level geometric premises discussed at the end.

For a weighted area (v) , let $(P_{\{B\}}(v))$ and $(P_{\{C\}}(v))$ be the sets of weighted perimeters of all admissible candidates of the indicated type having area exactly (v) , and define

[$I_{\{B\}}(v) = P_{\{B\}}(v)$, $I_{\{C\}}(v) = P_{\{C\}}(v)$.]

The admissible parameter domains and formulas are specified below. On the domain $(v > 0)$ used here, both sets are nonempty singletons; thus these definitions neither select a favorable branch nor depend on the value assigned to an empty infimum.

LEAN REFERENCES

```
CMVBallCase.lean CMVBallCase.typeBPerimeterValuesAtArea;
CMVBallCase.typeCPerimeterValuesAtArea; CMVBallCase.typeBPerimeterAtArea;
CMVBallCase.typeCPerimeterAtArea; CMVBallCase.exists_typeBRadius_at_area;
CMVBallCase.exists_typeCRadius_at_area;
CMVBallCase.typeBPerimeterAtArea_eq_radius;
CMVBallCase.typeCPerimeterAtArea_eq_radius
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Theorem. Let $(0 < \lambda < 1)$ and $(v_0 < v)$. If

$$[I_{\{C\}}(v_0) I_{\{B\}}(v_0),]$$

then

$$[I_{\{C\}}(v) < I_{\{B\}}(v).]$$

Thus weak dominance of type (C) at one such area becomes strict dominance at every larger area.

This is the formal persistence theorem, not an unconditional classification of isoperimetric regions.

LEAN REFERENCES

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CMVBallCase.lean CMVBallCase.typeC_dominance_persists
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Candidate formulas and global parameters

Fix $(0 < \lambda < 1)$ throughout the proof, and put

$$[a, s, S(t) = t - \lambda t.]$$

For a unit circle, $(S(t))$ is the area of the circular segment with central angle $(2t)$, for $(0 < t < \pi)$. For type (B), take

$$[0 < -a, = +a, r = .]$$

The two boundary arcs have lengths $(2r(-))$ and (2) . The region contains the unit disk. Its Euclidean area is the sum of the two circular segments cut by their common chord; subtracting $((1 - \lambda))$ gives its weighted area. Consequently the formulas in [CMV, p. 34, Figure 19 on p. 35] are

[

$$P_B(\beta) = 2((\pi - \beta)r + \lambda\alpha),$$

$$A_B(\beta) = r^2(\pi - \beta + \sin\beta\cos\beta) + S(\alpha) - (1 - \lambda)\pi.$$

]

The endpoint (=a) is retained: it has (r=0) and represents the unit disk. The endpoint () printed in the source is an infinite-radius limit, not a finite candidate.

LEAN REFERENCES

```
CMVBallCase.lean CMVBallCase.BAdmissible; CMVBallCase.typeBRadius_endpoint_zero;
                  CMVBallCase.arcSegment
CMVBallDefinitions.lean CMVBallCase.typeBAlpha; CMVBallCase.typeBRadius;
                        CMVBallCase.typeBPerimeter; CMVBallCase.typeBArea
```

For type (C), let $(0 < b < \pi/2)$, and set $(R=b)$. The inner and outer boundary arcs have lengths $(2Rb)$ and $(2R(-b))$. Orthogonality gives an overlap with the unit disk consisting of segments of areas $(R^2 S(b))$ and $(S(\pi/2 - b))$. Hence [CMV, p. 35]

[

$$P_C(b) = 2R(\pi - (1 - \lambda)b),$$

$$A_C(b) = R^2(\pi - (1 - \lambda)S(b)) - (1 - \lambda)S(\pi/2 - b).$$

]

These are exactly the source's formulas with $(R=1/b)$. Both angular endpoints are excluded degenerations. The elementary geometric descriptions explain the formulas; the formal objects are the explicit real-valued formulas, not sets equipped with a formalized perimeter measure.

LEAN REFERENCES

```
CMVBallCase.lean CMVBallCase.CAdmissible
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                        CMVBallCase.typeCArea
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Use radius as a global parameter. The identity and inverse charts are

[$r=+s$, $_ (R)=-$, $b(R)=1R$.]

The first chart is a bijection between the finite type-(B) angle domain and (R) ; the second is a bijection between $(0 < b < \pi/2)$ and $(R > 0)$. The radius maps are strictly decreasing in their angle parameters, and the displayed inverses work on their entire domains. In particular, the type-(B) chart includes $(R <)$, where $(_ (R) > \pi/2)$, as well as (R) ; no quadrant is discarded.

LEAN REFERENCES

```
CMVBallCase.lean CMVBallCase.typeBRadius_eq_add_sqrt_mul_cot; CMVBallCase.betaAtRadius;
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CMVBallCase.typeCRadius_strictAntiOn; CMVBallCase.betaAtRadius_admissible;
CMVBallCase.typeBRadius_betaAtRadius; CMVBallCase.betaAtRadius_typeBRadius;
CMVBallCase.bAtRadius_admissible; CMVBallCase.typeCRadius_bAtRadius;
CMVBallCase.bAtRadius_typeCRadius
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Write $(P_{\{B\}}(R), A_{\{B\}}(R))$ for the type-(B) formulas evaluated at $((R))$, and $(P_{\{C\}}(R), A_{\{C\}}(R))$ for the type-(C) formulas evaluated at $(b(R))$. Thus

[

$$\begin{aligned} P_B(R) &= 2 \left((\pi - \beta_\lambda(R))R + \lambda(\beta_\lambda(R) + a) \right), \\ A_B(R) &= R^2 \left(\pi - S(\beta_\lambda(R)) \right) + S(\beta_\lambda(R) + a) - (1 - \lambda)\pi, \\ P_C(R) &= 2R \left(\pi - (1 - \lambda)b(R) \right), \\ A_C(R) &= R^2 \left(\pi - (1 - \lambda)S(b(R)) \right) - (1 - \lambda)S(\pi/2 - b(R)). \end{aligned}$$

]

The radius formulas agree with the angular formulas for every admissible radius.

LEAN REFERENCES

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CMVBallCase.lean CMVBallCase.typeBPerimeterAtRadius; CMVBallCase.typeBAreaAtRadius;
CMVBallCase.typeCPerimeterAtRadius; CMVBallCase.typeCAreaAtRadius;
CMVBallCase.typeBPerimeterAtRadius_agrees;
CMVBallCase.typeBAreaAtRadius_agrees;
CMVBallCase.typeCPerimeterAtRadius_agrees;
CMVBallCase.typeCAreaAtRadius_agrees
```

Area and perimeter monotonicity

Differentiation, using $(S'(t)=2^2t)$, gives, for $(R>0)$,

[

$$\begin{aligned} P_B'(R) &= 2 \left(\pi - S(\beta_\lambda(R)) \right), & A_B'(R) &= R P_B'(R), \\ P_C'(R) &= 2 \left(\pi - (1 - \lambda)S(b(R)) \right), & A_C'(R) &= R P_C'(R). \end{aligned}$$

]

Both perimeter derivatives are positive. Indeed, the first factor is $(S'(-_)(R)) > 0$, while $(0 < S(b(R)) < b(R) < 2)$ makes the second factor positive. Thus area and perimeter are strictly increasing in radius: on $([0,))$ for type (B), and on $((0,))$ for type (C). Continuity at the type-(B) endpoint supplies the closed endpoint assertion.

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CMVBallCase.arcSegment_lt_self;
CMVBallCase.typeBPerimeterAtRadius_hasDerivAt;
CMVBallCase.typeCPerimeterAtRadius_hasDerivAt;
CMVBallCase.typeBAreaAtRadius_hasDerivAt;
CMVBallCase.typeCAreaAtRadius_hasDerivAt;
CMVBallCase.typeB_perimeter_factor_pos;
CMVBallCase.typeC_perimeter_factor_pos;
CMVBallCase.typeBAreaAtRadius_strictMonoOn;
CMVBallCase.typeCAreaAtRadius_strictMonoOn;
CMVBallCase.typeBPerimeterAtRadius_strictMonoOn;
CMVBallCase.typeCPerimeterAtRadius_strictMonoOn
```

The relevant area ranges follow without an arbitrary inverse choice. For type (B),

$$[A_{\{B\}}(0)=, P_{\{B\}}(0)=2, A_{\{B\}}(R) > 2R^2-(R).]$$

For type (C), the useful bounds are

$$[R^2 < A_{\{C\}}(R) < R^2 (R > 0).]$$

In particular, $(A_{\{C\}}(v) <^2 <)$, and its lower bound tends to infinity. The intermediate value theorem and strict monotonicity therefore give a unique positive radius of each type at every area $(v >)$. The inverse angle charts show that every admissible angular candidate of that area has this radius and this angle. Consequently the all-candidate perimeter sets are singletons and the infima are their actual perimeter values. Moreover, $(I_{\{B\}})$ is strictly increasing on $((,))$, and $(I_{\{B\}}(v) > 2)$ there.

LEAN REFERENCES

```
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CMVBallCase.typeBPerimeterAtArea_eq_radius;
CMVBallCase.typeCPerimeterAtArea_eq_radius;
CMVBallCase.typeBPerimeterAtArea_strictMonoOn;
CMVBallCase.typeBPerimeterAtArea_gt_commonEndpoint
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The fixed-radius comparison

We next prove the decisive inequality

$$[P_{\{C\}}(R) < P_{\{B\}}(R) (R > 0).]$$

This is a comparison at equal radii, not at equal areas. Half the perimeter difference is

$$[H(R) = (-R)_-(R) + (1-R)b(R).]$$

For $(0 < R)$, the first term is nonnegative and the other two are positive, so $(H(R) > 0)$.

LEAN REFERENCES

```
CMVBallCase.lean CMVBallCase.fixedRadiusHalfGap;
CMVBallCase.perimeter_gap_eq_two_mul_halfGap;
CMVBallCase.fixedRadiusHalfGap_pos_of_le
```

For the remaining region $(R > 1)$, vary the density while keeping (R) fixed. For $(0 < c < 1)$, define

[

$$\begin{aligned} D(c, R) &= (R - c)^2 + 1 - c^2, \\ L(c, R) &= \beta_c(R) - b(R) - \frac{c\sqrt{1-c^2}}{D(c, R)}, \\ N(c, R) &= R^3 - 4R^2c + 2Rc^2 + 3R - 2c. \end{aligned}$$

]

Here $(D(c, R) > 0)$, and direct differentiation yields

$$[c = R\{c^2\}L(c, R), (c, R) = -.]$$

The polynomial has the particularly useful derivative and diagonal value

$$[(c,R) = -(4(R-)^2 + 1 + 4R(1-c)) < 0 \ (R > 0, c < 1), N(R,R) = R(1-R^2).]$$

Thus $(N(R))$ is strictly decreasing on $((-1,1])$.

LEAN REFERENCES

```
CMVBallCase.lean CMVBallCase.fixedRadiusD; CMVBallCase.fixedRadiusL; CMVBallCase.fixedRadiusN;
CMVBallCase.fixedRadiusD_pos;
CMVBallCase.fixedRadiusHalfGap_density_ratio_hasDerivAt;
CMVBallCase.fixedRadiusL_density_hasDerivAt;
CMVBallCase.fixedRadiusN_density_derivative_identity;
CMVBallCase.fixedRadiusN_density_derivative_neg;
CMVBallCase.fixedRadiusN_diagonal; CMVBallCase.fixedRadiusN_strictAntiOn
```

We claim $(L(c,R) < 0)$ for $(0 < c < (1,R))$. First, $(L(0,R) = 0)$, and $(L(R))$ is continuous at zero. If $(N(c,R))$, then $(N(t,R) > 0)$ for $(0 < t < c)$. Hence (L) strictly decreases from zero on $([0,c])$, proving the claim, including the case $(N(c,R) = 0)$. If (R) , this case always applies, because

$$[N(c,R) > N(R,R) = R(1-R^2).]$$

If instead $(N(c,R) < 0)$, necessarily $(R > 1)$. For every $(c < t < 1)$, we have $(N(t,R) < 0)$, so (L) is strictly increasing to its left limit at one. The reciprocal angle formula

$$[_t(R) = (t < R)]$$

shows that this limit is $(-b(R) < 0)$. Therefore $(L(c,R) - b(R) < 0)$ in this case as well. The restriction $(c < R)$ is essential to this argument.

LEAN REFERENCES

```
CMVBallCase.lean CMVBallCase.fixedRadiusL_zero; CMVBallCase.fixedRadiusL_continuousAt_zero;
CMVBallCase.fixedRadiusL_neg_of_N_nonneg;
CMVBallCase.fixedRadiusN_pos_of_radius_le_one;
CMVBallCase.betaAtRadius_eq_arctan_reciprocal;
CMVBallCase.fixedRadiusL_tendsto_one; CMVBallCase.fixedRadiusL_neg_of_N_neg;
CMVBallCase.fixedRadiusL_neg
```

It follows that $(H(c,R)/c)$ is strictly decreasing for $(0 < c < (1,R))$. If $(< R < 1)$, continuity at $(c=R)$ and the already proved case of equal density and radius give

$$[> R > 0.]$$

If (R) , then

$$[\lim_{c \rightarrow 0}]$$

For ($R > 1$) this follows from the reciprocal angle formula. For ($R = 1$), the bound ($0 < c(1)/2$) makes $((1-c)c(1))$, while the remaining terms tend to zero. Strict decrease and this limit imply $(H(R)/>0)$: choose an intermediate density ($<c<1$), whose ratio is nonnegative by the limit, and compare strictly with it. Together with the case (R), this proves the fixed-radius inequality for every ($R > 0$).

LEAN REFERENCES

```
CMVBallCase.lean CMVBallCase.fixedRadiusHalfGap_density_ratio_strictAnti0n;
CMVBallCase.fixedRadiusHalfGap_density_ratio_continuousAt;
CMVBallCase.fixedRadiusHalfGap_density_ratio_tendsto_one_of_gt;
CMVBallCase.fixedRadiusHalfGap_density_ratio_tendsto_one_at_one;
CMVBallCase.fixedRadiusHalfGap_density_ratio_tendsto_one;
CMVBallCase.fixedRadiusHalfGap_pos_of_lt_radius_lt_one;
CMVBallCase.fixedRadiusHalfGap_pos_of_one_le_radius;
CMVBallCase.fixedRadiusHalfGap_pos_of_gt;
CMVBallCase.sameRadius_typeB_gt_typeC
```

Common perimeter and persistence

For ($p > 2$), let ($r_{\{B\}}(p)$) and ($r_{\{C\}}(p)$) be the unique positive radii satisfying

$$[P_{\{B\}}(r_{\{B\}}(p)) = p, P_{\{C\}}(r_{\{C\}}(p)) = p.]$$

To see existence, note that ($P_{\{B\}}(0) = 2$), ($P_{\{C\}}(0) < 2$), and

$$[R < P_{\{C\}}(R) < P_{\{B\}}(R) (R > 0).]$$

Continuity supplies the required radii, and strict monotonicity supplies uniqueness. The same-radius comparison then implies

$$[r_{\{B\}}(p) < r_{\{C\}}(p):]$$

otherwise monotonicity of ($P_{\{C\}}$) would give ($p P_{\{C\}}(r_{\{B\}}(p)) < p$). These inverses are used only on the genuine open common-perimeter domain.

LEAN REFERENCES

```
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CMVBallCase.existsUnique_typeBRadius_at_perimeter;
CMVBallCase.existsUnique_typeCRadius_at_perimeter;
CMVBallCase.typeBRadiusAtPerimeter; CMVBallCase.typeCRadiusAtPerimeter;
CMVBallCase.typeBRadiusAtPerimeter_lt_typeCRadiusAtPerimeter
```

Define the area difference at common perimeter by

$$[G(p)=A_{\{B\}}(r_{\{B\}}(p)) - A_{\{C\}}(r_{\{C\}}(p)).]$$

The perimeter derivatives are continuous and positive at positive radii, so the one-dimensional inverse function theorem applies locally. Using $(A_{\{B\}}'=R P_{\{B\}}')$ and $(A_{\{C\}}'=R P_{\{C\}}')$, the chain rule gives

$$[A_{\{B\}}(r_{\{B\}}(p))=r_{\{B\}}(p), A_{\{C\}}(r_{\{C\}}(p))=r_{\{C\}}(p).]$$

Consequently

$$[G'(p)=r_{\{B\}}(p)-r_{\{C\}}(p)<0.]$$

The mean value theorem on each compact subinterval of $((2,))$ makes (G) strictly decreasing throughout that domain. This is the monotone quantity used in the proof; no monotonicity of $(I_{\{B\}}-I_{\{C\}})$ is assumed.

LEAN REFERENCES

```
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CMVBallCase.typeBPerimeterAtRadius_hasStrictDerivAt;
CMVBallCase.typeCPerimeterAtRadius_hasStrictDerivAt;
CMVBallCase.commonPerimeterAreaGap_local_hasDerivAt;
CMVBallCase.commonPerimeterAreaGap_strictAnti
```

Now suppose $(<v_0<v)$ and $(I_{\{C\}}(v_0)I_{\{B\}}(v_0))$. Put $(p_0=I_{\{B\}}(v_0))$. At this perimeter the type-(B) area is (v_0) . The type-(C) radius realizing area (v_0) has perimeter at most (p_0) , so monotonicity of its perimeter and area maps gives

$$[A_{\{C\}}(r_{\{C\}}(p_0))v_0,G(p_0).]$$

Because $(I_{\{B\}})$ is strictly increasing, $(p=I_{\{B\}}(v)>p_0)$. Hence

$$[G(p)<G(p_0), A_{\{C\}}(r_{\{C\}}(p))>v.]$$

The unique type-(C) radius of area (v) is therefore strictly less than $(r_{\{C\}}(p))$; its perimeter is strictly less than $(p=I_{\{B\}}(v))$. This proves the theorem.

LEAN REFERENCES

```
CMVBallCase.lean CMVBallCase.commonPerimeterAreaGap_at_typeBProfile_nonpos;
                  CMVBallCase.typeCPerimeterAtArea_lt_of_commonPerimeterAreaGap_neg;
                  CMVBallCase.typeC_dominance_persists
```

Geometric scope

The verified result is persistence of type-(C) dominance over the faithful all-candidate type-(B)/(C) profiles. The formal argument establishes the full finite parameter domains, equality with the source formulas, existence and uniqueness at the required areas and perimeters, and the strict common-perimeter comparison. It does not itself establish existence or regularity of finite-perimeter minimizers, their connectedness, or the exhaustiveness of the geometric candidate classification. Nor does the persistence implication alone assert that a crossing exists.

The paper-level conclusion ($v_1=v_2$), proposed in the remark after [CMV, Theorem 3.23, p. 34], follows only when this analytic theorem is combined with the finite-perimeter existence, regularity, connectedness, and exhaustive candidate-classification premises underlying Proposition 3.22 and Theorem 3.23, together with the latter theorem’s small- and large-area alternatives. Under those external premises, above () every minimizer is of type (B) or (C), type (B) wins near that endpoint, and type (C) wins for sufficiently large areas. The continuous candidate profiles therefore meet; at their meeting both types are optimal, and persistence excludes any later return of type (B). Thus the transition thresholds can be taken equal. These geometric-measure-theory premises and this global minimizer classification are not formalized in this package; they are distinct from the verified analytic profile theorem.

Bibliography

[CMV] Antonio Cañete, Michele Miranda Jr., and Davide Vittone, *Some isoperimetric problems in planes with density*, arXiv:0906.1256v1 [math.DG], 8 June 2009. Relevant locators: Proposition 3.22, pp. 32–33; Theorem 3.23 and its following remark, pp. 34–36; candidate formulas and Figure 19, pp. 34–35.